better scientific software

Better Software for Science with High-Performance Computing

Presented by

PESO RAPIDS

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CASS

Consortium for the Advancement of Scientific Software

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With prior support from

ECP EXASCALE COMPUTING PROJECT IDEAS productivity

Anshu Dubey and Akash Dhruv

SupercomputingAsia 2026 (SCA 2026)/The International Conference on High Performance Computing in Asia-Pacific Region 2026 (HPCAsia 2026) conference

Slides, late-breaking updates, etc. at:
https://bssw-tutorial.github.io/
and click the link for today's tutorial

License, Citation and Acknowledgements

License and Citation

- **The requested citation the overall tutorial is:** Anshu Dubey and Akash Dhruv, Better Software for Science with High-Performance Computing tutorial, in SupercomputingAsia 2026 (SCA 2026)/The International Conference on High Performance Computing in Asia-Pacific Region 2026 (HPCAsia 2026), Osaka, Japan, 2026. DOI: 10.6084/m9.figshare.31130062.
- Individual modules may be cited as *Speaker, Module Title*, in *Tutorial Title*, …

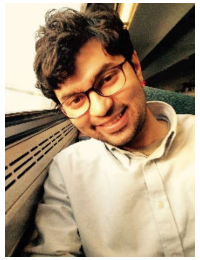
Acknowledgements

- This work was supported by the U.S. Department of Energy, Office of Science, Office of Advanced Scientific Computing Research, Next-Generation Scientific Software Technologies (NGSST) and Scientific Discovery through Advanced Computing (SciDAC) programs.
- This work was supported by the U.S. Department of Energy Office of Science, Office of Advanced Scientific Computing Research (ASCR), and by the Exascale Computing Project (17-SC-20-SC), a collaborative effort of the U.S. Department of Energy Office of Science and the National Nuclear Security Administration.
- This work was performed in part at the Argonne National Laboratory, which is managed by UChicago Argonne, LLC for the U.S. Department of Energy under Contract No. DE-AC02-06CH11357.
- This work was performed in part at the Lawrence Livermore National Laboratory, which is managed by Lawrence Livermore National Security, LLC for the U.S. Department of Energy under Contract No. DE-AC52-07NA27344.
- This work was performed in part at the Los Alamos National Laboratory, which is managed by Triad National Security, LLC for the U.S. Department of Energy under Contract No.89233218CNA000001
- This work was performed in part at the Oak Ridge National Laboratory, which is managed by UT-Battelle, LLC for the U.S. Department of Energy under Contract No. DE-AC05-00OR22725.
- This work was performed in part at Sandia National Laboratories. Sandia National Laboratories is a multi-mission laboratory managed and operated by National Technology and Engineering Solutions of Sandia, LLC., a wholly owned subsidiary of Honeywell International, Inc., for the U.S. Department of Energy's National Nuclear Security Administration under contract DE-NA0003525.

About Us

- Anshu Dubey, ANL
- Akash Dhruv, ANL

Anshu
she/her



Akash
he/him

- Participants in the Consortium for the Advancement of Scientific Software (CASS) and DOE/ASCR software stewardship projects: https://cass.community/
  - Members of the RAPIDS project
- Past involvement in
  - Exascale Computing Project
  - IDEAS Productivity family of projects: https://ideas-productivity.org/
  - And more!

Building an Online Community

https://bssw.io

- **New community-based resource for scientific software improvement**
- A central hub for sharing information on practices, techniques, experiences, and tools to improve developer productivity and software sustainability for computational science & engineering (CSE)

Goals

- Raise awareness of the importance of **good software practices** to scientific productivity and to the quality and reliability of computationally-based scientific results
- Raise awareness of the **increasing challenges** facing CSE software developers as high-end computing heads to extreme scales
- Help CSE researchers **increase effectiveness** as well as leverage and impact
- **Facilitate CSE collaboration via software** in order to advance scientific discoveries

Site users can…

- **Find information** on scientific software topics
- **Contribute new resources** based on your experiences
- Create content tailored to the unique needs and perspectives of a focused scientific domain

Follow BSSw

- BSSw Digest: https://bssw.io/pages/receive-our-email-digest
  - Updates on BSSw content
    - New blog posts, events, and resources
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  - Typically 1-2 messages per month
  - Also: RSS feed: https://bssw.io/items.rss

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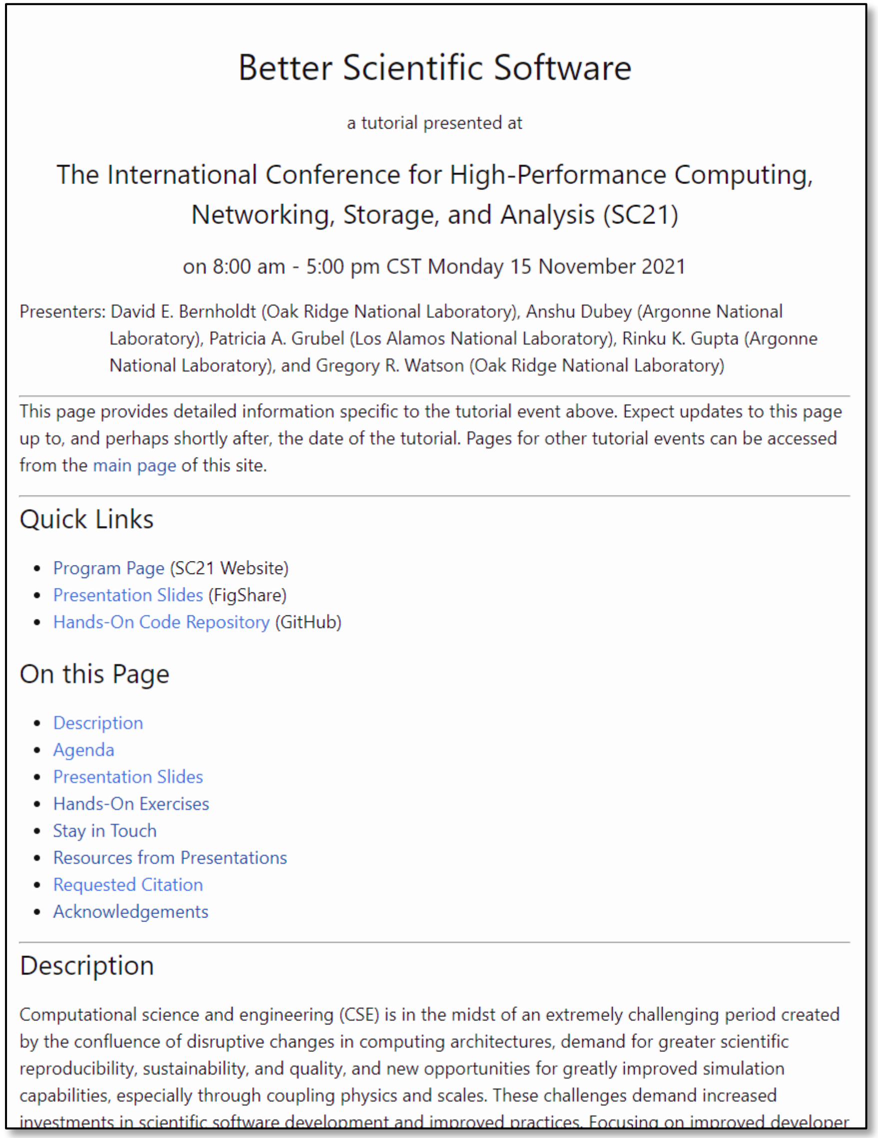
BSSw Tutorial Web Site

- https://bssw-tutorial.github.io/ is the one URL you need to find all the resources for this tutorial
- Each tutorial event has its own page
- Each tutorial page is considered archival
  - All of the materials used in that tutorial (or links to them)
  - Materials may be updated if we find mistakes



Better Scientific Software

a tutorial presented at

The International Conference for High-Performance Computing, Networking, Storage, and Analysis (SC21)

on 8:00 am - 5:00 pm CST Monday 15 November 2021

Presenters: David E. Bernholdt (Oak Ridge National Laboratory), Anshu Dubey (Argonne National Laboratory), Patricia A. Grubel (Los Alamos National Laboratory), Rinku K. Gupta (Argonne National Laboratory), and Gregory R. Watson (Oak Ridge National Laboratory)

This page provides detailed information specific to the tutorial event above. Expect updates to this page up to, and perhaps shortly after, the date of the tutorial. Pages for other tutorial events can be accessed from the main page of this site.

Quick Links

- Program Page (SC21 Website)
- Presentation Slides (FigShare)
- Hands-On Code Repository (GitHub)

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Description

Computational science and engineering (CSE) is in the midst of an extremely challenging period created by the confluence of disruptive changes in computing architectures, demand for greater scientific reproducibility, sustainability, and quality, and new opportunities for greatly improved simulation capabilities, especially through coupling physics and scales. These challenges demand increased investments in scientific software development and improved practices. Focusing on improved developer

Explaining Slide 2

- Slide 2 in all of our presentations contains the license, citation, and acknowledgements for the tutorial
- (Software) best practice to make your license and preferred citation(s) easily findable
- Sponsor acknowledgements rarely hurt!

License, Citation and Acknowledgements

License and Citation

- This work is licensed under a Creative Commons Attribution 4.0 International License (CC BY 4.0).
- **The requested citation the overall tutorial is:** Anshu Dubey, David E. Bernholdt, Better Scientific Software tutorial, in ISC High Performance, Hamburg, Germany and online, 2024

Acknowledgements

- Material included in these presentation is derived from work supported by the U.S. Department of Energy Office of Science, Office of Advanced Scientific Computing Research (ASCR), and by the Exascale Computing Project (17-SC-20-SC), a collaborative effort of the U.S. Department of Energy Office of Science and the National Nuclear Security Administration.
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We Want to Interact with You!

- We find these tutorials most interesting and informative (for everyone) if you ask questions and share experiences!
  - We learn too!
- Please raise your hand at any time to ask a question
- After the tutorial email us at bssw-tutorial@lists.mcs.anl.gov
  - With questions or feedback
  - The list moderator will allow your messages to be posted
- Refer to bssw-tutorial.github.io page for all tutorial materials

Hands-On Activities

- **For detailed information, see the tutorial web page: https://bssw-tutorial.github.io/2026-01-26-hpcasia/**
- The hands-on session uses Code-Scribe
  - You will be using an LLM through its API interface
- Participation in the hands-on is encouraged but not required
  - The instructor's prompts and generated code will be made available to participants after the exercise
- **To participate in the hands-on, you will need Code-Scribe and access to a LLM chat tool**

Agenda

The agenda is also available on the tutorial web page. Visit **https://bssw-tutorial.github.io** and click on the link for today's tutorial

Time	Title	Presenter
9:30 AM	Introduction	Anshu Dubey (ANL)
9:45 AM	Motivation and Overview of Best Practices in HPC Software Development	Anshu Dubey (ANL)
10:15 AM	Code Translation from Fortran to C++ using CodeScribe	Akash Dhruv (ANL)
10:45 AM	*Morning break*	
11:15 AM	Scientific Software Design	Anshu Dubey (ANL)
11:50 AM	Software Testing and Verification	Anshu Dubey (ANL)
12:30 PM	*Lunch break*	
1:30 PM	Refactoring	Anshu Dubey (ANL)
2:00 PM	Improving Reproducibility Through Better Software Practices	Akash Dhruv (ANL)
2:45 PM	*Afternoon break*	
3:15 PM	Hands on coding tasks with LLMs	Akash Dhruv (ANL) and Anshu Dubey (ANL)
4:30 PM	*Adjourn*	